

DeepSeek-V4 Runtime and Speculative Decoding

Deep Dive into DeepSeek — Chapter 21

Yin Yang

Foundation Books Project

The one rule of this chapter

A V4 request is **not** determined by checkpoint weights alone.

The serving path carries one *committed-prefix boundary*.

- Admission fixes the model artifact, decoding policy, prompt length, and output reserve.

The one rule of this chapter

A V4 request is **not** determined by checkpoint weights alone.

The serving path carries one *committed-prefix boundary*.

- Admission fixes the model artifact, decoding policy, prompt length, and output reserve.
- Prefill builds compressed long-context state and exact local state.

The one rule of this chapter

A V4 request is **not** determined by checkpoint weights alone.

The serving path carries one *committed-prefix boundary*.

- Admission fixes the model artifact, decoding policy, prompt length, and output reserve.
- Prefill builds compressed long-context state and exact local state.
- Decode may add an optional draft path — but only a target-verified prefix may advance emitted text *and* every persistent state object.

The one rule of this chapter

A V4 request is **not** determined by checkpoint weights alone.

The serving path carries one *committed-prefix boundary*.

- Admission fixes the model artifact, decoding policy, prompt length, and output reserve.
- Prefill builds compressed long-context state and exact local state.
- Decode may add an optional draft path — but only a target-verified prefix may advance emitted text *and* every persistent state object.

Goal: follow one request through admission, prefill, draft, verify, and commit — and keep capacity separate from retrieval, draft proposals separate from committed state, and offline acceptance separate from production latency.

Admission, budgets, and cache accounting

Draft, verify, commit

Four draft families, four boundaries

DeepSpec: the public workflow

Attention state and the memory headline

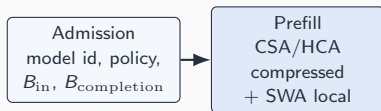
Evidence and what comes next

Admission, budgets, and cache accounting

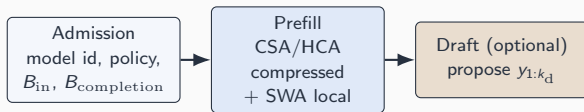
One request, one committed state

Admission
model id, policy,
 B_{in} , $B_{\text{completion}}$

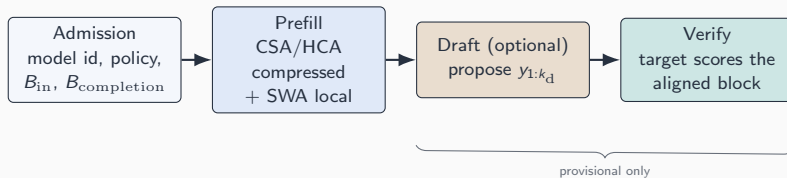
One request, one committed state



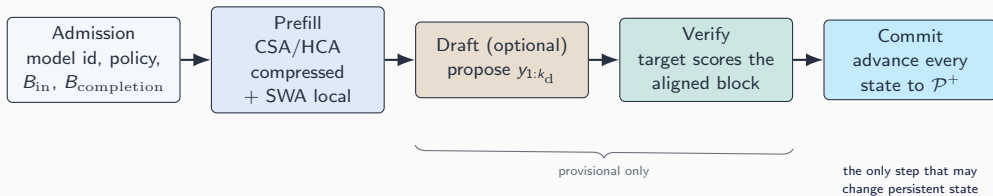
One request, one committed state



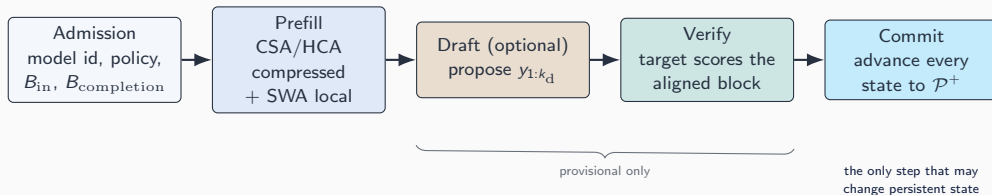
One request, one committed state



One request, one committed state

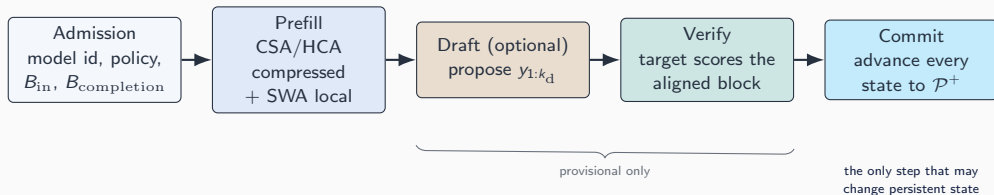


One request, one committed state



- Every stage is a separate obligation with a separate failure mode.
- The chapter's running workload is the CaseBridge long-document policy probe.

One request, one committed state



- Every stage is a separate obligation with a separate failure mode.
- The chapter's running workload is the CaseBridge long-document policy probe.

The final evaluation record must separate acceptance behavior from latency, throughput, resource use, and answer quality.

Capacity is arithmetic; retrieval is a separate test

Definition (Context window)

The maximum number of input and generated tokens admitted under a cited release. With context length L_{ctx} and planned budgets $B_{\text{in}}, B_{\text{completion}}$, feasibility is $B_{\text{in}} + B_{\text{completion}} \leq L_{\text{ctx}}$.

Budget segment	Tokens	Why it is reserved
Input packet	920,000	documents, statutes, notes, instructions
Completion reserve	60,000	total generated-token room
Slack	20,000	tokenizer variance, headers, schema fields

Capacity is arithmetic; retrieval is a separate test

Definition (Context window)

The maximum number of input and generated tokens admitted under a cited release. With context length L_{ctx} and planned budgets $B_{\text{in}}, B_{\text{completion}}$, feasibility is $B_{\text{in}} + B_{\text{completion}} \leq L_{\text{ctx}}$.

Budget segment	Tokens	Why it is reserved
Input packet	920,000	documents, statutes, notes, instructions
Completion reserve	60,000	total generated-token room
Slack	20,000	tokenizer variance, headers, schema fields
Total planned window	1,000,000	$920000 + 60000 + 20000$

Capacity is arithmetic; retrieval is a separate test

Definition (Context window)

The maximum number of input and generated tokens admitted under a cited release. With context length L_{ctx} and planned budgets $B_{\text{in}}, B_{\text{completion}}$, feasibility is $B_{\text{in}} + B_{\text{completion}} \leq L_{\text{ctx}}$.

Budget segment	Tokens	Why it is reserved
Input packet	920,000	documents, statutes, notes, instructions
Completion reserve	60,000	total generated-token room
Slack	20,000	tokenizer variance, headers, schema fields
Total planned window	1,000,000	$920000 + 60000 + 20000$

The one-million-token limit establishes request capacity only. [REPORTED: DeepSeek V4 preview release note]

Five things the word “cache” names here

Meaning	What it explains
context-cache fields	prompt-token reuse in the response usage row
serving KV cache	attention state for an active request

Five things the word “cache” names here

Meaning	What it explains
context-cache fields serving KV cache	prompt-token reuse in the response usage row attention state for an active request
MLA latent cache	compressed per-position attention information
V4 heterogeneous cache	CSA, HCA, indexer, sliding-window, tail states
on-disk KV storage	reuse of selected internal cache objects

Five things the word “cache” names here

Meaning	What it explains
context-cache fields serving KV cache	prompt-token reuse in the response usage row attention state for an active request
MLA latent cache	compressed per-position attention information
V4 heterogeneous cache	CSA, HCA, indexer, sliding-window, tail states
on-disk KV storage	reuse of selected internal cache objects

When the interface reports a prompt total with its components, the ledger closes:

$$T_{\text{prompt}} = T_{\text{hit}} + T_{\text{miss}}.$$

Five things the word “cache” names here

Meaning	What it explains
context-cache fields serving KV cache	prompt-token reuse in the response usage row attention state for an active request
MLA latent cache	compressed per-position attention information
V4 heterogeneous cache	CSA, HCA, indexer, sliding-window, tail states
on-disk KV storage	reuse of selected internal cache objects

When the interface reports a prompt total with its components, the ledger closes:

$$T_{\text{prompt}} = T_{\text{hit}} + T_{\text{miss}}.$$

A measurement note must name its cache meaning before drawing any conclusion about memory, latency, or retrieval quality.

Draft, verify, commit

The commit rule, in three positions

Position	Draft	Target	Verifier action	Committed prefix
1	a	a	accept proposal	$\mathcal{P}a$
2	b	b	accept proposal	$\mathcal{P}ab$

The commit rule, in three positions

Position	Draft	Target	Verifier action	Committed prefix
1	a	a	accept proposal	$\mathcal{P}a$
2	b	b	accept proposal	$\mathcal{P}ab$
3	c	x	reject c ; emit x	$\mathcal{P}abx$

The commit rule, in three positions

Position	Draft	Target	Verifier action	Committed prefix
1	a	a	accept proposal	$\mathcal{P}a$
2	b	b	accept proposal	$\mathcal{P}ab$
3	c	x	reject c ; emit x	$\mathcal{P}abx$

- Greedy acceptance runs while y_j equals the target's greedy choice; at the first mismatch the target token is emitted and the rest discarded.
- Stochastic decoding needs an exact acceptance-and-correction rule that preserves the target distribution — greedy equality is only the deterministic special case.

The commit rule, in three positions

Position	Draft	Target	Verifier action	Committed prefix
1	a	a	accept proposal	$\mathcal{P}a$
2	b	b	accept proposal	$\mathcal{P}ab$
3	c	x	reject c ; emit x	$\mathcal{P}abx$

- Greedy acceptance runs while y_j equals the target's greedy choice; at the first mismatch the target token is emitted and the rest discarded.
- Stochastic decoding needs an exact acceptance-and-correction rule that preserves the target distribution — greedy equality is only the deterministic special case.

Three serial target decisions became one block verification. That does not make target computation free.

From concept to code: only the accepted prefix is committed

The public evaluator builds its committed extension from exactly two pieces: the accepted draft prefix and one target token.

```
from DeepSpec/deepspec/eval/base_evaluator.py

committed_tokens = torch.cat(
    [
        proposal.verify_input_ids[:, 1 : accepted_draft_tokens + 1],
        next_token.unsqueeze(1),
    ],
    dim=1,
)
```

From concept to code: only the accepted prefix is committed

The public evaluator builds its committed extension from exactly two pieces: the accepted draft prefix and one target token.

```
from DeepSpec/deepspec/eval/base_evaluator.py

committed_tokens = torch.cat(
    [
        proposal.verify_input_ids[:, 1 : accepted_draft_tokens + 1],
        next_token.unsqueeze(1),
    ],
    dim=1,
)
```

The slice stops at `accepted_draft_tokens`: no state derived only from the rejected suffix can reach the committed prefix. Everything past that index was provisional work. [INSPECTED: DeepSpec

`eval/base_evaluator.py`]

Four draft families, four boundaries

Not interchangeable speedup labels

Method	Main idea	V4 interaction
HyperDFlash	mHC-aware draft state, gated residual reduction	preserves expanded residual state before reduction
DFlash	one-pass block-diffusion drafting on target hidden features	public DeepSpec targets are Qwen3 and Gemma4; no released V4 draft
EAGLE-3	multi-layer feature fusion, direct token prediction, training-time test	public configs for Qwen3 and Gemma4; no V4 EAGLE-3 artifact

Not interchangeable speedup labels

Method	Main idea	V4 interaction
HyperDFlash	mHC-aware draft state, gated residual reduction	preserves expanded residual state before reduction
DFlash	one-pass block-diffusion drafting on target hidden features	public DeepSpec targets are Qwen3 and Gemma4; no released V4 draft
EAGLE-3	multi-layer feature fusion, direct token prediction, training-time test	public configs for Qwen3 and Gemma4; no V4 EAGLE-3 artifact
DSpark	semi-autoregressive drafting plus confidence-scheduled verification	released Pro/Flash artifacts attach DSpark modules

Not interchangeable speedup labels

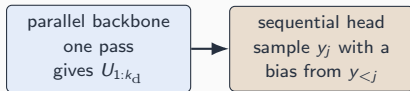
Method	Main idea	V4 interaction
HyperDFlash	mHC-aware draft state, gated residual reduction	preserves expanded residual state before reduction
DFlash	one-pass block-diffusion drafting on target hidden features	public DeepSpec targets are Qwen3 and Gemma4; no released V4 draft
EAGLE-3	multi-layer feature fusion, direct token prediction, training-time test	public configs for Qwen3 and Gemma4; no V4 EAGLE-3 artifact
DSpark	semi-autoregressive drafting plus confidence-scheduled verification	released Pro/Flash artifacts attach DSpark modules

None of these paths changes the base CSA/HCA, mHC, DeepSeekMoE, MTP, or Muon mechanism definitions. Speculative decoding is an optional serving layer. [REPORTED: method papers and released model cards]

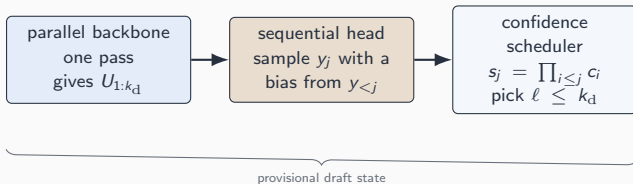
DSpark: build wide, stitch cheaply, verify what is worth it

parallel backbone
one pass
gives $U_{1:k_d}$

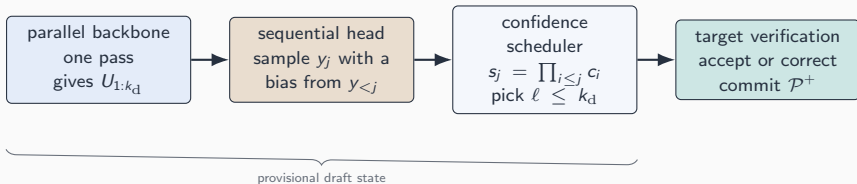
DSpark: build wide, stitch cheaply, verify what is worth it



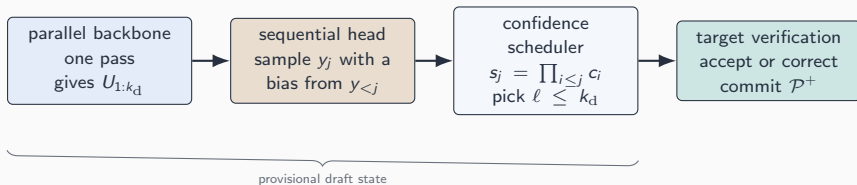
DSpark: build wide, stitch cheaply, verify what is worth it



DSpark: build wide, stitch cheaply, verify what is worth it

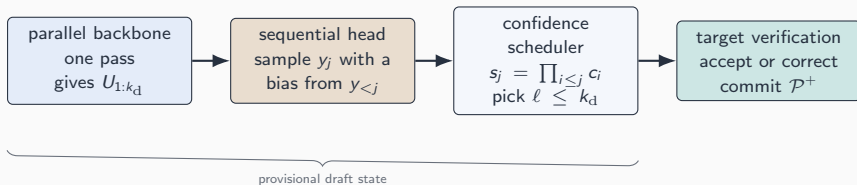


DSpark: build wide, stitch cheaply, verify what is worth it



- c_j estimates conditional survival, s_j estimates prefix survival, and ℓ is the scheduled verification width. **None of them commits a token.**
- Under load, sending a low-survival suffix to the target spends batch slots that could advance other requests.

DSpark: build wide, stitch cheaply, verify what is worth it



- c_j estimates conditional survival, s_j estimates prefix survival, and ℓ is the scheduled verification width. **None of them commits a token.**
- Under load, sending a low-survival suffix to the target spends batch slots that could advance other requests.

The paper reports live-traffic evaluation in V4 preview serving against an MTP-1 baseline.

[REPORTED: DSpark paper]

DeepSpec: the public workflow

A staged pipeline that stops before production

Stage	What it produces
prompt preparation	normalized conversations, held-out evaluation prompts, target-regenerated assistant turns
target cache	captured intermediate layers plus last hidden state, binary shards, dense index, versioned manifest
draft training	block proposals with Markov and confidence heads, gradients accumulated across ranks

A staged pipeline that stops before production

Stage	What it produces
prompt preparation	normalized conversations, held-out evaluation prompts, target-regenerated assistant turns
target cache	captured intermediate layers plus last hidden state, binary shards, dense index, versioned manifest
draft training	block proposals with Markov and confidence heads, gradients accumulated across ranks
checkpoint and resume	draft weights, resolved config, per-rank optimizer and RNG state, atomically retargeted pointer
offline evaluation	batch-size-one draft/verify replay with acceptance diagnostics

A staged pipeline that stops before production

Stage	What it produces
prompt preparation	normalized conversations, held-out evaluation prompts, target-regenerated assistant turns
target cache	captured intermediate layers plus last hidden state, binary shards, dense index, versioned manifest
draft training	block proposals with Markov and confidence heads, gradients accumulated across ranks
checkpoint and resume	draft weights, resolved config, per-rank optimizer and RNG state, atomically retargeted pointer
offline evaluation	batch-size-one draft/verify replay with acceptance diagnostics

Public scope ends at batch-size-one offline replay. Request admission, continuous batching, distributed KV state, and throughput-aware verification scheduling belong to production serving. [INSPECTED: deepseek-ai/DeepSpec pinned commit]

Acceptance diagnostics are not a speedup

Evaluator output	Companion measurement it still needs
<code>draft_tokens_per_proposal</code> <code>acceptance_length</code>	timed runtime: kernel latency, verifier cost, batch capacity subtract the extra target token to get draft-only accepted length

Acceptance diagnostics are not a speedup

Evaluator output	Companion measurement it still needs
<code>draft_tokens_per_proposal</code> <code>acceptance_length</code>	timed runtime: kernel latency, verifier cost, batch capacity subtract the extra target token to get draft-only accepted length
<code>verify_rate</code>	wall-clock timing and request throughput
<code>accept_rate@j</code>	a task scorer for correctness or answer quality

Acceptance diagnostics are not a speedup

Evaluator output	Companion measurement it still needs
<code>draft_tokens_per_proposal</code> <code>acceptance_length</code>	timed runtime: kernel latency, verifier cost, batch capacity subtract the extra target token to get draft-only accepted length
<code>verify_rate</code>	wall-clock timing and request throughput
<code>accept_rate@j</code>	a task scorer for correctness or answer quality

- A high mean accepted length can still *reduce* throughput if feature extraction, verification, or synchronization consumes more target capacity than it saves.
- A modest accepted length can help under a light-load latency objective.

Acceptance diagnostics are not a speedup

Evaluator output	Companion measurement it still needs
<code>draft_tokens_per_proposal</code> <code>acceptance_length</code>	timed runtime: kernel latency, verifier cost, batch capacity subtract the extra target token to get draft-only accepted length
<code>verify_rate</code>	wall-clock timing and request throughput
<code>accept_rate@j</code>	a task scorer for correctness or answer quality

- A high mean accepted length can still *reduce* throughput if feature extraction, verification, or synchronization consumes more target capacity than it saves.
- A modest accepted length can help under a light-load latency objective.

So the measurement must name its load profile and scheduler policy.

Attention state and the memory headline

Runtime ownership, not compression math

Resident object	Runtime obligation	Failure at the boundary
CSA compressed KV & indexer	finalize complete groups; keep indices attached to what they name	a stale index selects a legal but wrong entry
HCA compressed KV	append completed non-overlapping groups	group boundaries or prefix length disagree
SWA state	retain the exact window and the tail needed for compression	eviction removes a still-readable position

Runtime ownership, not compression math

Resident object	Runtime obligation	Failure at the boundary
CSA compressed KV & indexer	finalize complete groups; keep indices attached to what they name	a stale index selects a legal but wrong entry
HCA compressed KV	append completed non-overlapping groups	group boundaries or prefix length disagree
SWA state	retain the exact window and the tail needed for compression	eviction removes a still-readable position
Committed-prefix record	advance every state surface to one accepted position	rollback leaves one surface ahead of the emitted text

Runtime ownership, not compression math

Resident object	Runtime obligation	Failure at the boundary
CSA compressed KV & indexer	finalize complete groups; keep indices attached to what they name	a stale index selects a legal but wrong entry
HCA compressed KV	append completed non-overlapping groups	group boundaries or prefix length disagree
SWA state	retain the exact window and the tail needed for compression	eviction removes a still-readable position
Committed-prefix record	advance every state surface to one accepted position	rollback leaves one surface ahead of the emitted text

The architecture chapter defines the transformations. The runtime asks which state each configured layer owns, when it becomes readable, and what a rollback must restore.

What the “98% less memory” headline measures

Comparison at 1M context	FLOPs/token (FP8-equiv.)	KV-cache size	Source support
V4-Pro vs. V3.2	27%	10%	report summary
V4-Flash vs. V3.2	about 10%	about 7%	ratios read from Figure 1

[REPORTED: DeepSeek-V4 technical report]

What the “98% less memory” headline measures

Comparison at 1M context	FLOPs/token (FP8-equiv.)	KV-cache size	Source support
V4-Pro vs. V3.2	27%	10%	report summary
V4-Flash vs. V3.2	about 10%	about 7%	ratios read from Figure 1
V4 series vs. BF16 GQA8, head dim 128	not given	about 2%	attention-cache comparison

[REPORTED: DeepSeek-V4 technical report]

What the “98% less memory” headline measures

Comparison at 1M context	FLOPs/token (FP8-equiv.)	KV-cache size	Source support
V4-Pro vs. V3.2	27%	10%	report summary
V4-Flash vs. V3.2	about 10%	about 7%	ratios read from Figure 1
V4 series vs. BF16 GQA8, head dim 128	not given	about 2%	attention-cache comparison

[REPORTED: DeepSeek-V4 technical report]

- The headline 98% is the complement of that last row's $\approx 2\%$ retained KV-cache size.
- Total GPU memory also holds weights, activations, and workspaces, so it does not inherit the reduction. The 27% entry is FP8-equivalent single-token FLOPs, not latency.

What the “98% less memory” headline measures

Comparison at 1M context	FLOPs/token (FP8-equiv.)	KV-cache size	Source support
V4-Pro vs. V3.2	27%	10%	report summary
V4-Flash vs. V3.2	about 10%	about 7%	ratios read from Figure 1
V4 series vs. BF16 GQA8, head dim 128	not given	about 2%	attention-cache comparison

[REPORTED: DeepSeek-V4 technical report]

- The headline 98% is the complement of that last row's $\approx 2\%$ retained KV-cache size.
- Total GPU memory also holds weights, activations, and workspaces, so it does not inherit the reduction. The 27% entry is FP8-equivalent single-token FLOPs, not latency.

Percentages in different rows compare only after you keep the named baseline and metric.

Evidence and what comes next

The row that is still empty

Question	Public basis	Production details still needed
Context and cache accounting	reported capacity plus derived state equations	admission rule, layout, hit/miss ownership, memory peaks
Draft attachment	method papers and released draft modules	target/draft revisions, feature attachment, runtime route
Offline acceptance	DeepSpec evaluator and replay artifacts	workload, parser, loss mask, seed, target-call count

[OPEN: production V4 serving trace]

The row that is still empty

Question	Public basis	Production details still needed
Context and cache accounting	reported capacity plus derived state equations	admission rule, layout, hit/miss ownership, memory peaks
Draft attachment	method papers and released draft modules	target/draft revisions, feature attachment, runtime route
Offline acceptance	DeepSpec evaluator and replay artifacts	workload, parser, loss mask, seed, target-call count
Production speedup	no complete public V4 production trace	scheduler, load profile, latency percentiles, throughput, quality acceptance

[OPEN: production V4 serving trace]

The row that is still empty

Question	Public basis	Production details still needed
Context and cache accounting	reported capacity plus derived state equations	admission rule, layout, hit/miss ownership, memory peaks
Draft attachment	method papers and released draft modules	target/draft revisions, feature attachment, runtime route
Offline acceptance	DeepSpec evaluator and replay artifacts	workload, parser, loss mask, seed, target-call count
Production speedup	no complete public V4 production trace	scheduler, load profile, latency percentiles, throughput, quality acceptance

[OPEN: production V4 serving trace]

Until those measurements exist, the public mechanisms and the offline workflow support a mechanism-level result only.

Concept-to-code map for the chapter

- **prefix-safe commit and exact acceptance** → `DeepSpec/deepspec/eval/base_evaluator.py`

Concept-to-code map for the chapter

- **prefix-safe commit and exact acceptance** → `DeepSpec/deepspec/eval/base_evaluator.py`
- **DSpark loss and the overlap identity** → `DeepSpec/deepspec/modeling/dspark/loss.py`
- **confidence-threshold early stop** → `DeepSpec/deepspec/eval/dspark/draft_ops.py`

Concept-to-code map for the chapter

- **prefix-safe commit and exact acceptance** → `DeepSpec/deepspec/eval/base_evaluator.py`
- **DSpark loss and the overlap identity** → `DeepSpec/deepspec/modeling/dspark/loss.py`
- **confidence-threshold early stop** → `DeepSpec/deepspec/eval/dspark/draft_ops.py`
- **pinned DSpark / DFlash / EAGLE-3 configs** → `DeepSpec/config/*/*_qwen3_4b.py`
- **atomic checkpoint pointer publication** → `DeepSpec/deepspec/utils/io.py`

Concept-to-code map for the chapter

- **prefix-safe commit and exact acceptance** → `DeepSpec/deepspec/eval/base_evaluator.py`
- **DSpark loss and the overlap identity** → `DeepSpec/deepspec/modeling/dspark/loss.py`
- **confidence-threshold early stop** → `DeepSpec/deepspec/eval/dspark/draft_ops.py`
- **pinned DSpark / DFlash / EAGLE-3 configs** → `DeepSpec/config/*/*_qwen3_4b.py`
- **atomic checkpoint pointer publication** → `DeepSpec/deepspec/utils/io.py`
- **target-cache shard and loss-token defaults** → `DeepSpec/scripts/data/prepare_target_cache.py`

Concept-to-code map for the chapter

- **prefix-safe commit and exact acceptance** → `DeepSpec/deepspec/eval/base_evaluator.py`
- **DSpark loss and the overlap identity** → `DeepSpec/deepspec/modeling/dspark/loss.py`
- **confidence-threshold early stop** → `DeepSpec/deepspec/eval/dspark/draft_ops.py`
- **pinned DSpark / DFlash / EAGLE-3 configs** → `DeepSpec/config/*/*_qwen3_4b.py`
- **atomic checkpoint pointer publication** → `DeepSpec/deepspec/utils/io.py`
- **target-cache shard and loss-token defaults** → `DeepSpec/scripts/data/prepare_target_cache.py`

Admit the request, keep every state surface on one committed prefix,
and never let an accepted draft stand in for a measured deployment.

Next: the FlashMLA chapter follows this attention state down into its concrete kernel interfaces.